

LEDGER · 2026-08-28 · MATERIALS SCIENCE / CARBON MINERALIZATION

Advanced Polymer Reinforcement via Brine-Mediated CO2 Mineralization of Aragonite Whiskers

REJECTED

This concept did **not** clear the framework.

The reasoning is published in full below.

What the assessment found

Incumbent benchmark	Granular Calcite / Ground Calcium Carbonate (GCC)
Measured advantage	Increases polymer tensile strength by up to 90% with a 24.0 aspect ratio vs GCC which reduces strength by 10-20%.
Time to first revenue	24 months
Regulatory risk	Moderate — qualification costs reported (\$500,000 estimated)

Why it failed

- Voided by: Illegitimate incumbent

Assessment

Ground Calcium Carbonate is a volumetric filler used for cost reduction rather than a structural reinforcement fiber, making it an illegitimate comparator for tensile strength claims. Additionally, Tier-1 automotive compounding validation requires over 18 months.

Also flagged

- Evidence rests on non-peer-reviewed sources
- no_production_runsheet

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-28 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

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